



Effect of irradiation on corrosion of 304 nuclear grade stainless steel in simulated PWR primary water



Ping Deng^{a,b}, Qunjia Peng^{a,c,*}, En-Hou Han^a, Wei Ke^a, Chen Sun^c, Zhijie Jiao^d

^a Key Laboratory of Nuclear Materials and Safety Assessment, Institute of Metal Research, Chinese Academy of Sciences, Shenyang City, 110016, China

^b School of Materials Science and Engineering, University of Science and Technology of China, Shenyang City, 110016, China

^c State Power Investment Corporation Research Institute, Beijing, 102209, China

^d Department of Nuclear Engineering and Radiological Sciences, University of Michigan, Ann Arbor, MI, 48109, USA

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ABSTRACT

Effect of irradiation on corrosion of 304 nuclear grade stainless steel (304NG SS) in simulated primary water of pressurized water reactor was investigated. The investigation was conducted by comparing the microstructure of the oxide scale formed on the steel irradiated to different doses by proton. The results revealed that increasing irradiation dose promoted both the corrosion and intergranular corrosion of 304NG SS. This is attributed to the irradiation induced defects as well as the Cr depletion at the grain boundary in the steel.

1. Introduction

Irradiation-assisted stress corrosion cracking (IASCC) of austenitic stainless steel (SS) core components is one major concern for maintenance of nuclear power plants. The IASCC is a complex phenomenon as a result of the synergic effect of irradiation induced damage of the material, corrosion and stress [1–4]. Studies showed that increasing irradiation dose resulted in a higher degree of intergranular cracking [1,5–7]. The irradiation induced damage of the material includes the formation of dislocation loops, grain boundary segregation and hardening etc., which occur simultaneously at comparable rates and are all correlated with the increased cracking susceptibility [7–9]. Further, deformation of irradiated alloys led to dislocation channeling as a result of dislocation glide [4,10]. As such, deformation was localized into the channels in which dislocations were fed into the grain boundary region [11–13]. The change in deformation modes and the interaction of localized dislocation channels with the grain boundary promoted the IASCC [10–13].

To date, studies on the effect of irradiation on IASCC had mainly focused on the irradiation induced changes in microstructure and mechanical properties. The role of corrosion in IASCC, however, has not been paid sufficient attentions. According to the internal oxidation and slip-dissolution/oxidation mechanisms for stress corrosion cracking (SCC), oxide film rupture is the first step in the SCC process and thereby the corrosion behavior is crucial to SCC initiation [14,15]. In fact, a study reported that the irradiated SS that was resistant to intergranular cracking in 288 °C argon was susceptible to intergranular cracking in

high temperature water at the same temperature [11], suggesting an apparent effect of corrosion on IASCC. As such, attentions should also be paid on the corrosion behavior of irradiated alloys, in particular to the localized corrosion at grain boundary, in order to understand further the mechanism of IASCC.

In recent years, the effect of irradiation on corrosion in high temperature water has received increased attentions, while only a few studies have been reported. Jiao found no obvious acceleration of corrosion by post-irradiation in high temperature water [16], while Perrin et al. concluded oppositely [17,18]. The promotion of corrosion by in-situ proton irradiation was also reported by Wang et al. [19]. These studies suggest that the role of irradiation in corrosion has not been fully understood, particularly for the post-irradiation corrosion. Further, while studies by Perrin et al. concluded some effect of post-irradiation on corrosion of austenitic stainless steels in high temperature water [17,18], discussions on the effect of irradiation had been focused on the irradiation induced structural defects. The effect of the possible change in chemical composition by radiation-induced segregation (RIS) on corrosion, particularly to the localized corrosion at grain boundary, however, has received limited attentions. In addition, there is still a lack of detailed understanding of the quantitative dependence of the corrosion behavior of austenitic SS in high temperature water on irradiation doses.

This study focused on the effect of irradiation on corrosion of 304 nuclear grade (NG) SS in simulated primary water of pressurized water reactor (PWR). The objective is to correlate the irradiation dose to the overall corrosion, as well as the RIS to localized corrosion at the grain

* Corresponding author at: Key Laboratory of Nuclear Materials and Safety Assessment, Institute of Metal Research, Chinese Academy of Sciences, Shenyang City, 110016, China.
E-mail address: pengqunjia@yahoo.com (Q. Peng).

Table 1
Chemical composition (wt.%) of 304NG SS.

C	Mn	Si	S	P	Ni	Cr	Co	Fe
0.04	1.73	0.27	0.002	0.021	8.87	19.51	0.04	Bal.

boundary. Irradiation-induced structure of 304NG SS was characterized using transmission electron microscope (TEM). Corrosion behavior of the steel irradiated to various doses by proton was studied by exposure tests in simulated PWR primary water, followed by characterization of the oxide scale using scanning electron microscopy (SEM), X-ray photoelectron spectroscopy (XPS) and TEM.

2. Experimental

2.1. Material

The material used was solution annealed (SA) 304NG SS with a chemical composition listed in Table 1. Bar-type specimens with a dimension of 20 mm × 3 mm × 2 mm cut from the alloy were wet ground using SiC papers up to 3000 grit, and then polished by diamond pastes of 1.5- and 0.5-μm to obtain a mirror-like surface. After wet-polishing, the specimens were polished by 40-nm colloidal silica slurry for over 2 h to ensure the removal of residual strain prior to proton irradiation. Irradiations were conducted using the General Ionex Tandemron accelerator at the Ion Beam Laboratory of the University of Michigan. The specimens were irradiated with 2 MeV protons at 360 °C at a rate of approximately 6×10^{-6} displacement per atom (dpa) per second (dpa/s) to a dose of 0.5-, 1.5-, 3- and 5-dpa, respectively. The depth damage profile as calculated by the Stopping and Range of Ions in Materials 2008 (SRIM 2008) computer code is shown in Fig. 1 [20]. The damage profile is nearly uniform within the ~1 to ~10 μm of the whole 20 μm-thick damaged layer, and shows an implantation peak at ~18 μm. Details about the proton irradiation process are available elsewhere [12,21].

2.2. TEM characterization of the irradiation induced defects and segregation

The SA, 0.5- and 3-dpa irradiated specimens were selected for characterization of the irradiation induced defect and grain boundary segregation by TEM. The specimens were electropolished in an electrolyte at 30 °C and 30 mA for 5 s to remove a damage layer of 3–4 μm thick from the irradiated surface. The electrolyte was prepared by 5 vol. % perchloric acid of 70 wt.% concentration and 95% vol.% ethanol of 99 wt.% concentration. Then the non-irradiated face of the specimen was exposed for wet grinding to reduce the thickness to be less than

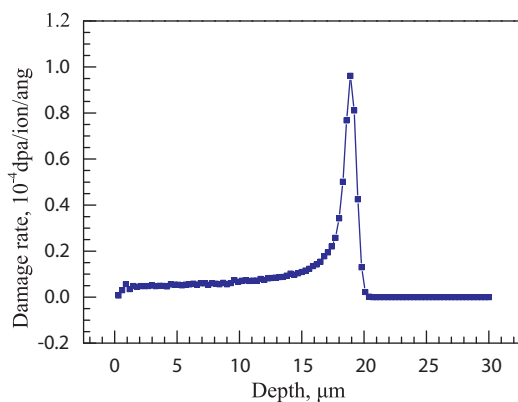


Fig. 1. Damage rate depth profile for 304NG SS irradiated with 2.0 MeV protons as calculated by SRIM 2008 computer code [20].

60 μm. TEM discs were then drilled out through the thinned specimens by a drill bit with a 3-mm inner diameter. These discs were thinned further from the non-irradiated side to electron transparency with a TenuPol-5 single-jet electropolisher, using the same electrolyte at –20 °C and 30 V. Following the electro-thinning, microstructural characterization was conducted using a JEM-2100 TEM equipped with an energy dispersive X-ray spectroscopy (EDX). Spot analyses with a probe beam diameter of 1 nm were carried out to determine composition profiles across high-angle grain boundaries (HAGBs). Prior to the measurement, the grain boundary was aligned edge-on to the electron beam, which minimized the projected width of the grain boundary during sample tilting. In total more than 5 EDX line scans across the grain boundary were performed to calculate composition profiles for Cr and Ni. Faulted Frank dislocation loops were imaged by the rel-rod technique [22], and about 500 loops were imaged for each irradiation condition to obtain average loop diameters and number densities. Since the rel-rod image showed faulted loops from only one of the four sets of (1 1 1) planes, the number of faulted loops measured must be multiplied by four to get the total number of faulted loops [12,22]. The total number of faulted loops then divided by the volume involved in the picture gave the number density of the faulted loops. Thickness of the local area where the dislocation loops were imaged was determined from the electron energy-loss spectrum (EELS) by measuring the ratio of the zero loss peak intensity to that of the total spectrum [23].

2.3. Exposure test

After irradiation, the specimens were polished again by 40-nm colloidal silica slurry for half an hour to get fresh surfaces, which removed the surface with a thickness of 1–2 μm and could ensure that the specimen surfaces following different doses of irradiation were consistent. Then the exposure testing was conducted at 320 °C in simulate PWR primary water in a 3-L autoclave made of 316 SS, which is equipped with a refreshed water loop. The water contained 1200 mg/L of B as H₃BO₃ and 2.3 mg/L of Li as LiOH·H₂O, with a dissolved hydrogen (DH) concentration of 2.6 mg/L at the inlet. The DH was achieved by applying a hydrogen overpressure of 0.08 MPa in the water tank of the loop, following removal of air in the water by bubbling hydrogen. To study the effect of irradiation dose on corrosion, in total four doses at 0.5, 1.5, 3 and 5 dpa as well as the SA specimens with no irradiation were employed for the exposure tests with a fixed exposure period of 500 h. After the exposure, the specimens were removed from the autoclave for characterization of the oxide scale.

2.4. Analysis of the oxide scale

Following the exposure tests, characterization of the oxide scales formed on the SA, 0.5-, 1.5-, 3- and 5-dpa specimens were performed to clarify the corrosion behavior. The surface morphology of the oxide scale was examined by a FEI XL30 Field Emission Gun SEM operating at 15 kV. The depth profile of the oxide scale was analyzed with an ESCALAB 250 XPS system. A selected area of 2-mm diameter was sequentially sputtered using a 2 keV Ar ion beam at a rate of 0.1 nm/s referred to Ta₂O₅ layer. More details about the XPS analysis were presented elsewhere [24]. Since XPS sputtering is a homogeneous process, the depth profiles consist of average element concentrations, which depend on the local structure of the oxide scale. Therefore, the recorded depth profiles were only used to estimate the thickness of the oxide scale as a function of irradiation dose. The accurate depth profile on the cross-section of the oxide scale was analyzed using a JEM-2100 TEM operating at 200 kV. The TEM is equipped with scanning capabilities for producing chemical contrast images and the EDX device for the composition analysis. Both point- and area-scans were used to analyze the chemical composition of the oxide scale. High-resolution TEM images were captured by a Gatan 4k charge coupled device camera and analyzed using Fast Fourier Transform (FFT) technique to

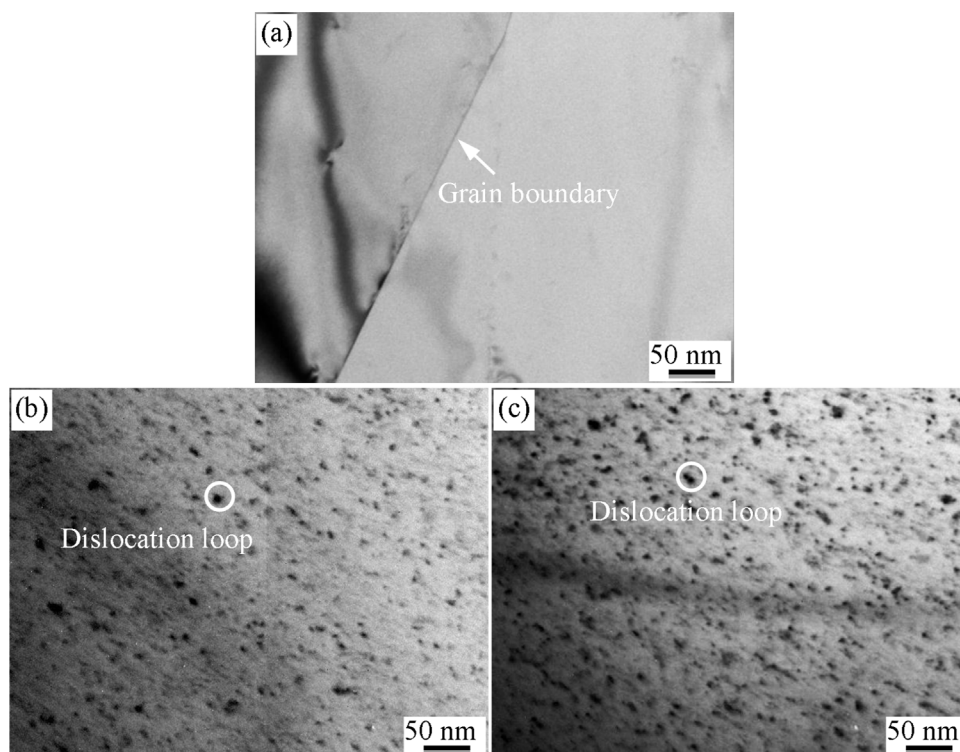


Fig. 2. TEM bright field observation of microstructures of (a) solution annealed, (b) 0.5-dpa and (c) 3-dpa irradiated specimens of 304NG SS on a $g = 200$, 2-beam condition ($z \sim [110]$).

study crystallographic details of the oxide. The oxide scales formed on the SA, 0.5- and 3-dpa specimens were selected for the TEM observation and analysis. The TEM samples of the cross-section of the oxide scale were prepared using a focused ion beam with Ga ion sputtering (FEI Quanta 200 3D). Each sample contained a grain boundary to allow the observation and analysis of the intergranular corrosion. The grain boundary selected for the sample is HAGB, which was identified by electron diffraction pattern of TEM before the TEM analysis. In total seven grain boundaries were analyzed, including three for the 0.5-dpa, three for the 3-dpa and one for the SA specimens.

3. Results

3.1. Irradiation induced defect and segregation in 304NG SS

TEM observation of the microstructure of the SA, 0.5-dpa and 3-dpa irradiated specimens is shown in Fig. 2. The irradiation resulted in formation of high density dislocation loops, while no such defects were observed in the SA specimen. Fig. 3 summarizes the observation results,

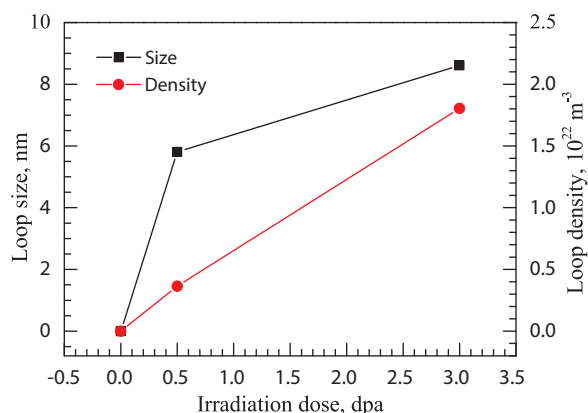


Fig. 3. Measured changes in density and size of dislocation loops as a function of irradiation dose.

which shows that the loop density was increased by a higher irradiation dose. Specifically, the density was in the order of 10^{-22} m^{-3} and the size was 6–10 nm, both of which are well consistent with the literature data [12,25]. Fig. 4 presents typical concentration profiles of Cr and Ni across the grain boundaries in the SA, 0.5-dpa and 3-dpa irradiated specimens. While no change in composition was observed at the grain boundary in the SA specimen, the RIS at the grain boundary in the irradiated specimens increased with increasing the irradiation dose.

3.2. Surface morphology of the oxide scale

SEM observation of the oxide scale formed on the SA, 0.5-, 1.5-, 3- and 5-dpa specimens following the exposure is shown in Fig. 5. The oxide scale consists of a layer of polyhedral oxide particles with a size of ~ 0.2 to $2 \mu\text{m}$ and the underneath continuous oxide film. Further, it is worthwhile to note that the number and size of the oxide particles were increased by a higher dose, suggesting a higher corrosion rate.

3.3. Depth profile of the oxide scale analyzed by XPS

XPS depth profile of O 1s in the oxide scale formed on the SA, 0.5-, 1.5-, 3- and 5-dpa specimens following the exposure is shown in Fig. 6(a). The thickness of the oxide scale is defined from the depth profile where the intensity of oxygen level decreases to 50% of the value at the surface [24,26], which was highlighted by dash lines. Fig. 6(b) summarizes the measured oxide scale thickness as a function of irradiation dose. As can be seen, the thickness of the oxide scale increased with increasing the irradiation dose, indicating the irradiation promoted corrosion of 304NG SS in PWR primary water. Precise analyses of the structure and composition of the oxide scale by TEM are presented in the following section.

3.4. Depth profile of the oxide scale analyzed by TEM

3.4.1. Overall corrosion behavior

Fig. 7 shows the TEM observation and analysis of the cross-sections of the oxide scale formed on the SA, 0.5-dpa and 3-dpa specimens. It

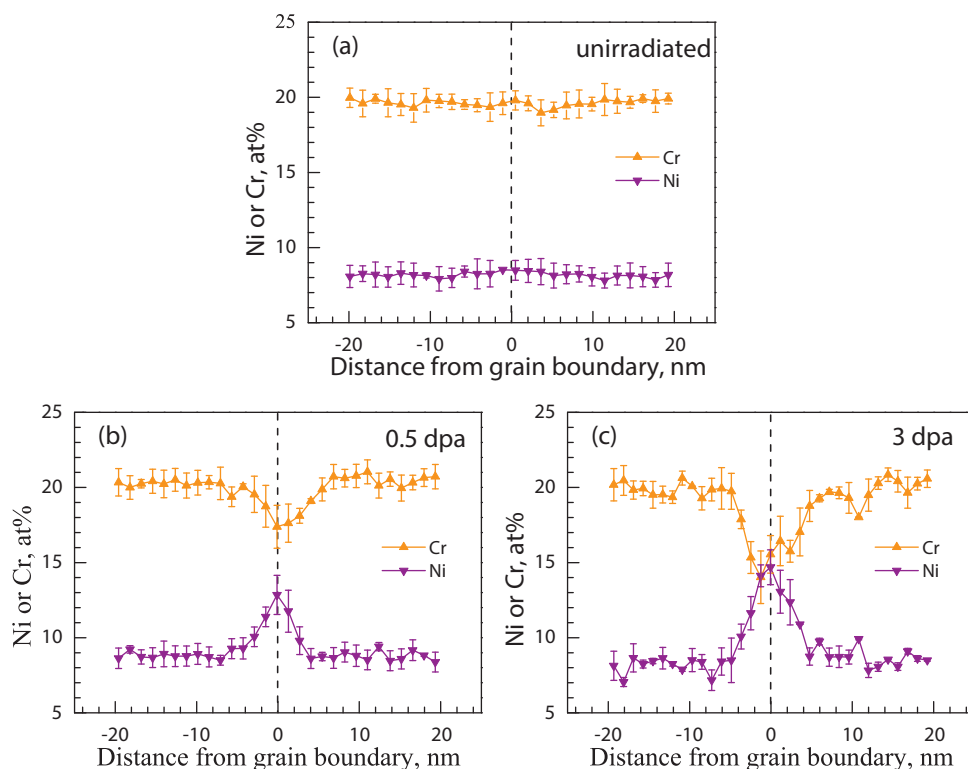


Fig. 4. Composition profiles across high-angle grain boundaries on (a) solution annealed, (b) 0.5-dpa and (c) 3-dpa irradiated specimens of 304NG SS.

revealed a similar duplex-type oxide on both SA and irradiated specimens as that observed by SEM, i.e. the outer layer of polyhedral oxide particles and the inner continuous oxide film. Further, both the EDX mappings and line scans revealed that the composition of the oxide formed on the SA, 0.5- and 3-dpa specimens was similar. The outer oxide particles mainly consist of Fe and Ni oxide, suggesting they were Fe-Ni spinel. The inner continuous oxide, on the other hand, consists of oxides of Ni, Cr and Fe, suggesting it was Fe-Cr-Ni spinel [24,27,28]. In addition, there is no obvious difference in Cr concentration of the inner oxide formed on the SA, 0.5-dpa and 3-dpa specimens.

Nevertheless, observation of the oxide scale also displayed differences in thickness and structure of the inner continuous oxide film. A thicker inner oxide film was observed to form at a higher irradiation dose (Fig. 7). This result as well as the SEM observation shown in Fig. 5 suggested that the irradiation promoted the overall corrosion of 304NG SS in PWR primary water. More detailed microstructure of the inner oxide film for the SA, 0.5-dpa and 3-dpa specimens was observed and analyzed by using high-resolution TEM (Fig. 8). The crystallographic

analysis of the oxide by FFT (equivalent to electron diffraction patterns) is shown in the insets in the figure. A weakened character of lattice was observed in the oxide film of the irradiated specimens, particularly to the 3-dpa specimen, while the pattern of the oxide film of the SA specimen showed a clear lattice character. As concluded by previous studies, the more speckled diffraction lattice implies a larger grain size of the spinel oxide [27,29]. As such, a finer oxide grain in the inner oxide film of the oxide scale on 3-dpa specimen can be expected. In fact, a decreased size of grains of the oxide formed on Xe-implanted alloy 690 in high temperature water has been observed [18].

3.4.2. Localized corrosion at grain boundary

The morphology and microstructure of the oxide formed at the grain boundary in SA, 0.5-dpa and 3-dpa specimens after the 500-h exposure were also observed and analyzed by TEM, shown in Figs. 9 through 11. It was found that hump-like oxide at the grain boundary was formed, suggesting the occurrence of localized corrosion. While the morphology of the oxide at the grain boundary was similar, the depth of the

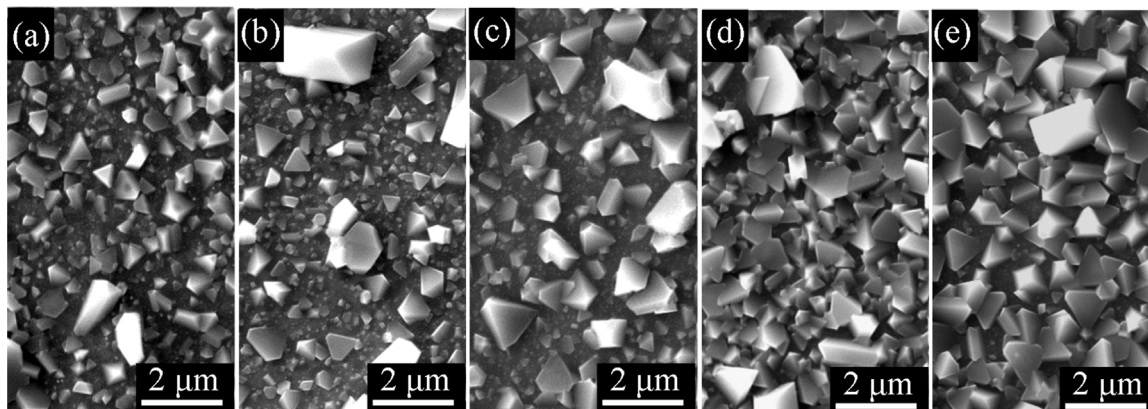


Fig. 5. SEM observation of surface morphology of the oxide scales formed on 304NG SS subjected to different irradiation doses following the 500-h exposure in simulated PWR primary water at 320 °C. (a) solution annealed, (b) 0.5 dpa, (c) 1.5 dpa, (d) 3 dpa, (e) 5 dpa.

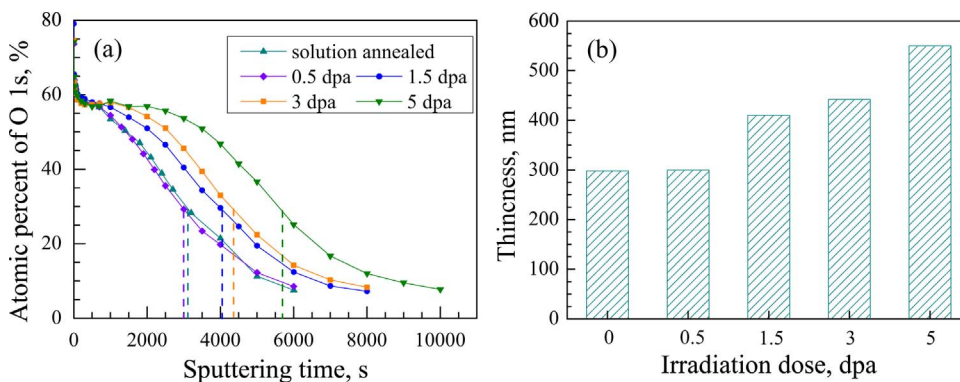


Fig. 6. (a), XPS depth profiles of O 1s in the oxide scales formed on solution annealed, 0.5-, 1.5-, 3- and 5-dpa irradiated specimens of 304NG SS following the 500-h exposure in simulated PWR primary water at 320 °C. The dashed lines highlighted the half-height of O 1s; (b), Thickness of the oxide scales on 304NG SS as a function of irradiation dose.

penetration oxidation through the surface oxide film into the grain boundary showed difference. On SA specimen, a penetration of oxidation with a depth of approximately 40 nm was observed, shown in Fig. 9(a). The oxide penetration at the grain boundary in 0.5-dpa and 3-dpa specimens, however, was much deeper. As shown in Figs. 10(a) and 11(a), increasing the irradiation dose from 0.5 to 3 dpa caused an

increase of the oxidation depth at the grain boundary from ~60 to ~120 nm. This suggested a promoted intergranular corrosion by irradiation. The crystallographic study by FFT shown in the insets in Figs. 9(a), 10(a) and 11(a) indicated that the intergranular oxide in the SA, 0.5-dpa and 3-dpa specimens had a similar diffraction character of amorphous phase.

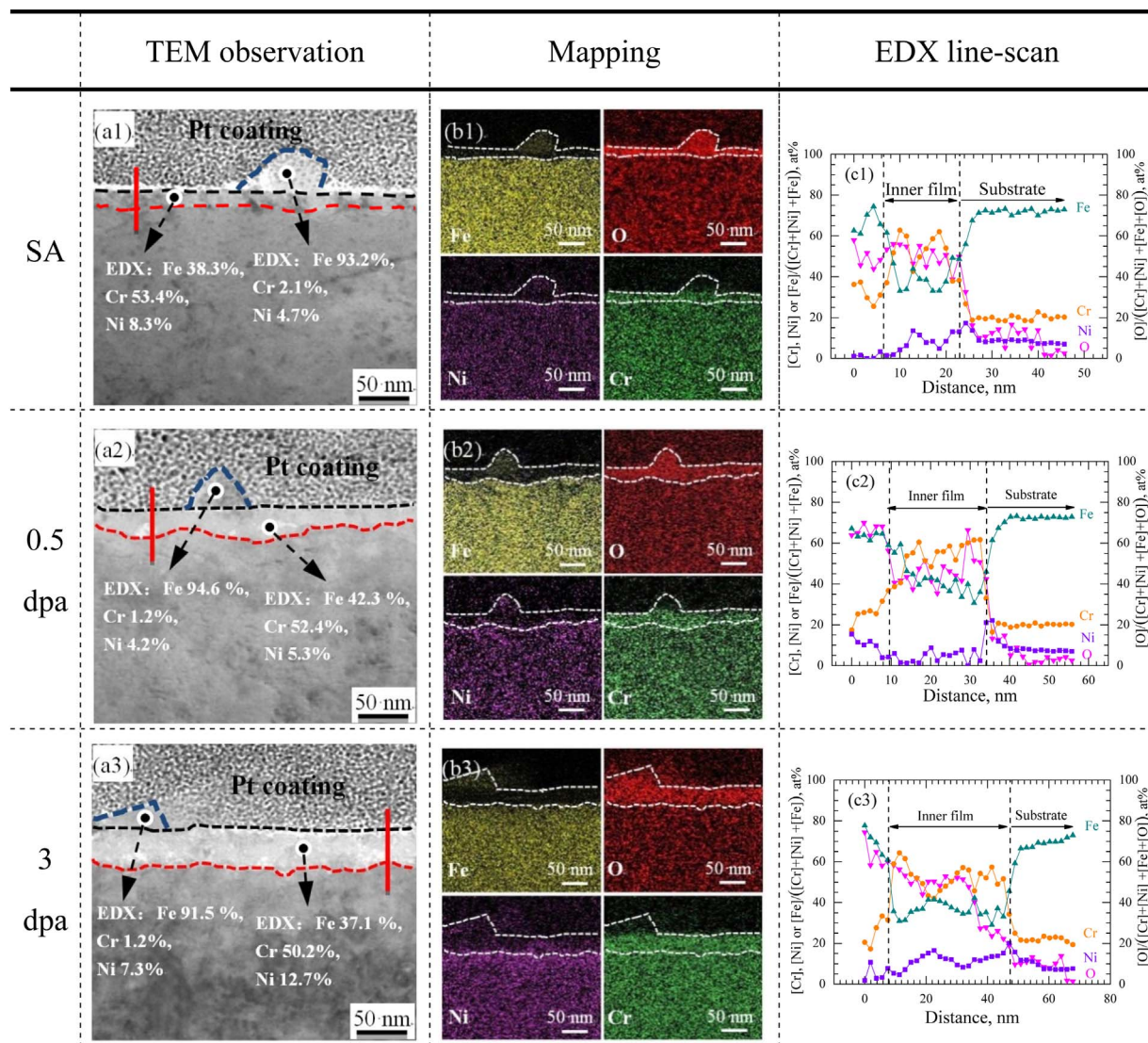


Fig. 7. TEM observation and analysis of the cross-section of the oxide scales formed on (a1, b1, c1) solution annealed, (a2, b2, c2) 0.5-dpa and (a3, b3, c3) 3-dpa irradiated specimens of 304NG SS following the 500-h exposure in simulated PWR primary water. a1, a2 and a3 are TEM observations of the cross-section of the oxide scales. b1, b2 and b3 are corresponding mappings for Fe, O, Cr and Ni. c1, c2 and c3 are point-scans collected along the red line shown in a1, a2 and a3, respectively.

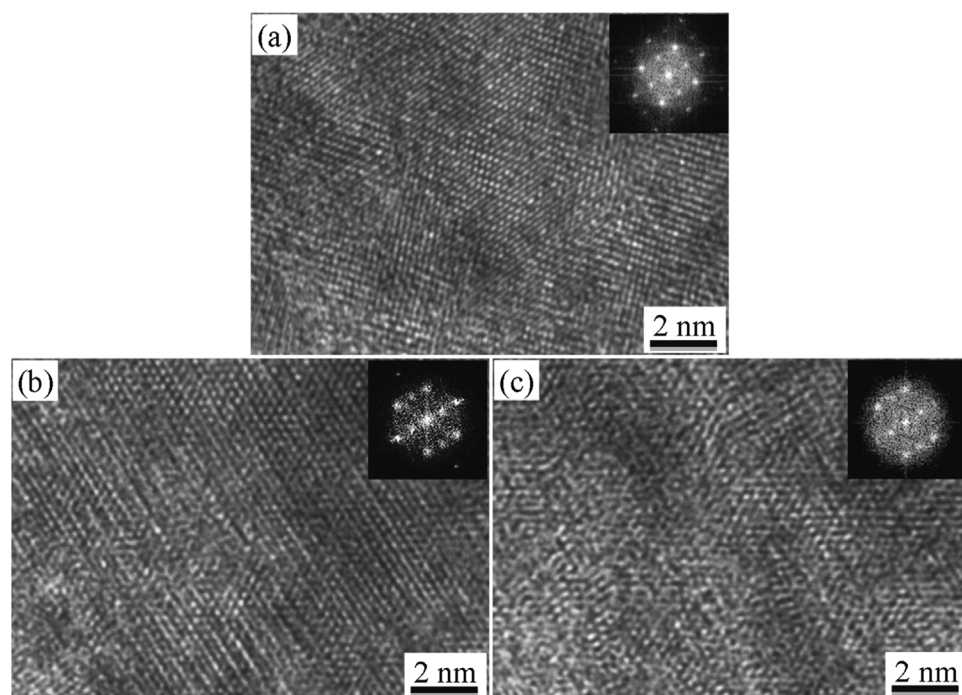


Fig. 8. High resolution TEM observation of the inner oxide film formed on (a) solution annealed, (b) 0.5-dpa and (c) 3-dpa irradiated specimens of 304NG SS following the 500-h exposure in simulated PWR primary water.

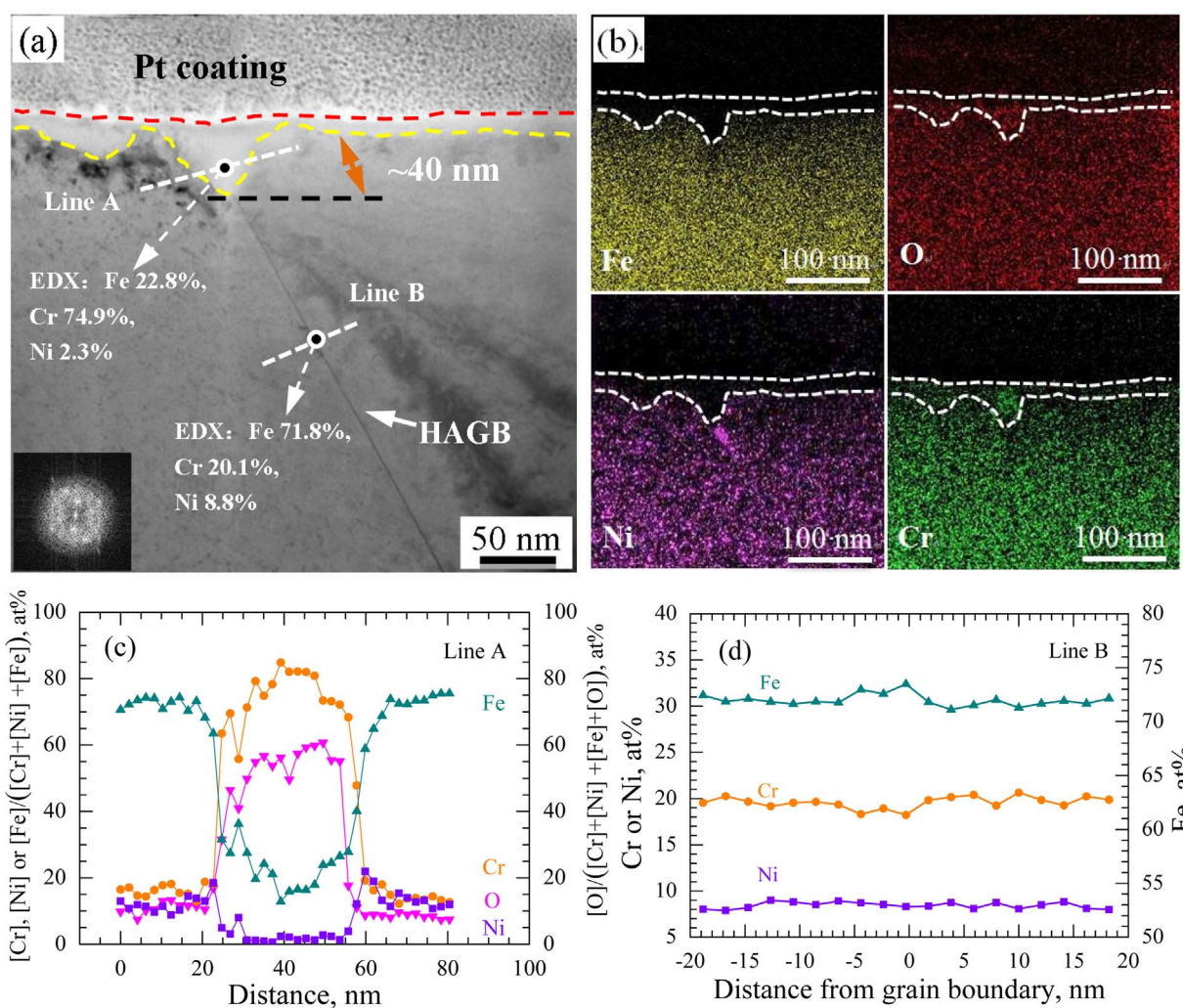


Fig. 9. (a), TEM observation of the localized corrosion at a grain boundary in solution annealed 304NG SS. (b), Mappings for Cr, Fe, Ni, and O showing oxidation along the grain boundary. (c) and (d), EDX point-scans collected along lines A and B shown in (a), respectively.

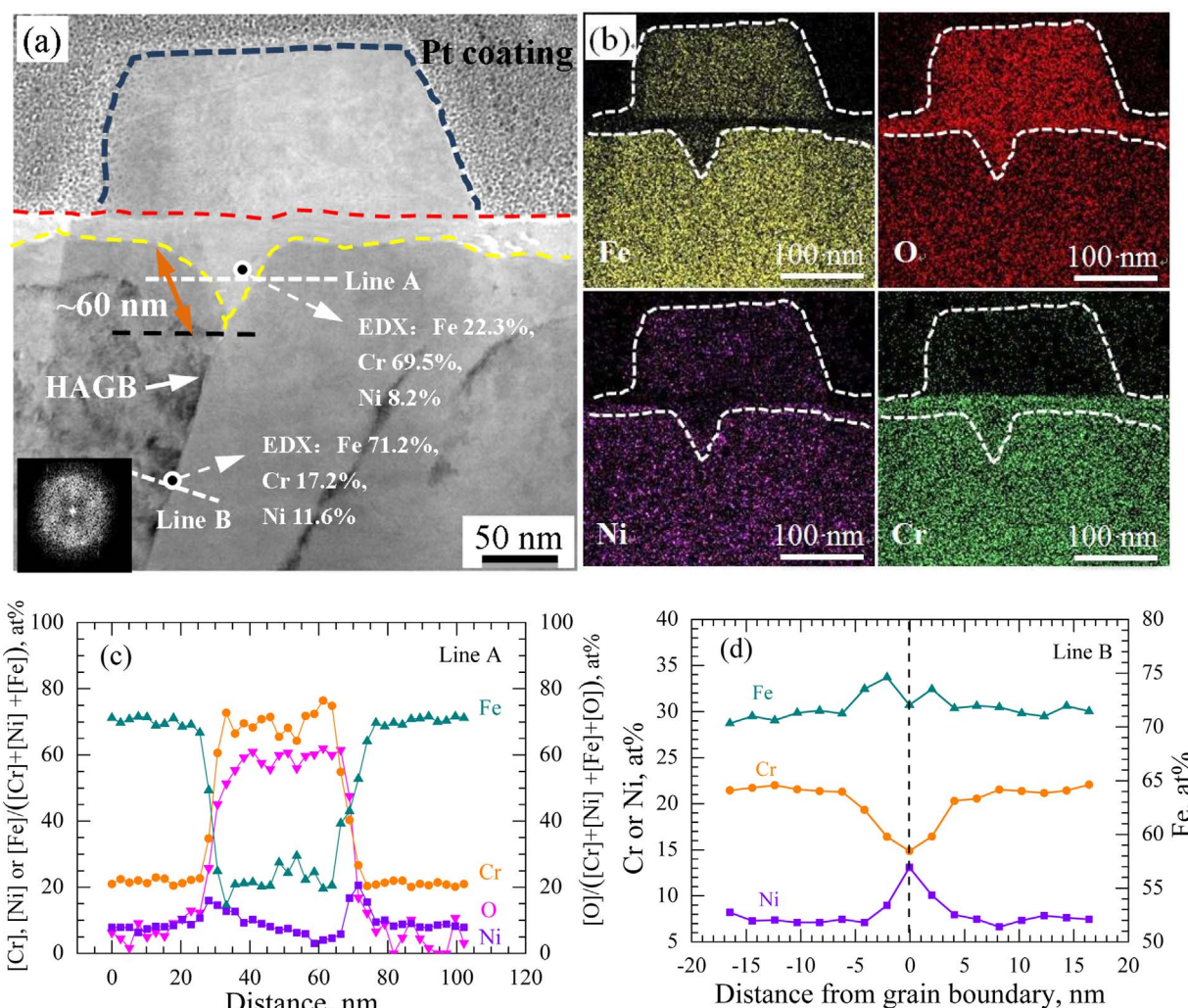


Fig. 10. (a), TEM observation of the localized corrosion at a grain boundary in 0.5-dpa irradiated 304NG SS. (b), Mappings for Cr, Fe, Ni, and O showing oxidation along the grain boundary. (c) and (d), EDX point-scans collected along lines A and B shown in (a), respectively.

EDX line-scans and mappings as shown in Figs. 9 through 11 revealed the composition profile of the oxide formed at the grain boundary in SA, 0.5-dpa and 3-dpa specimens. While the oxides were all depleted in Fe and enriched in Cr, the Cr content normalized by excluding the oxygen varied with the irradiation dose. It was 74.9 at.% on SA specimen, and decreased to 69.5 at.% by the 0.5-dpa irradiation and further to 58.3 at.% by the 3-dpa irradiation, shown in Figs. 9(a), 10(a) and 11 (a). Further, Ni enrichment ahead of the oxide at the grain boundary in SA specimen was observed, while no such a phenomenon for 0.5-dpa and 3-dpa specimens (Figs. 9 (b), 10(b) and 11(b)).

RIS at the grain boundary was also analyzed by EDX line-scans, shown in Figs. 9(d), 10(d) and 11(d). The composition profile across the grain boundary beneath the oxide film with a distance of over 100 nm in the 0.5- and 3-dpa specimens showed a depletion of Cr and an enrichment of Ni, while no such a segregation in the SA specimen was observed. The magnitude of depletion of Cr at the grain boundary on 0.5- and 3-dpa specimens was 3.5 ± 0.9 at.% and 9.2 ± 0.9 at.%, respectively.

Fig. 12 summarized results of the analysis on all the seven grain boundaries, which suggested that the irradiation promoted intergranular corrosion of 304NG SS in PWR primary water. This was in association with a decreased Cr content at the primary grain boundary in the irradiated specimens.

4. Discussion

4.1. Influence of irradiation on overall corrosion of 304NG SS

The previous SEM observation and XPS analysis showed that a thicker oxide scale with a higher density and a larger size of the discrete oxide particles in the outer layer was formed on irradiated specimens. Further, the TEM observation revealed that the thickness of the inner continuous oxide film was also increased by a higher irradiation dose. These results indicated that increasing irradiation dose led to a higher corrosion rate of 304NG SS in PWR primary water.

According to the SEM and TEM observations shown in Figs. 5 and 7, the thickness of the inner oxide film is much smaller than that of the outer layer consisting of the oxide particles. As such, the change in the size and density of the discrete oxide particles was most likely the decisive factor for the change in the total thickness of the oxide scale. The size and density of dislocation loops generated by irradiation were increased by a higher dose (Figs. 2 and 3). These loops are preferential sites for nucleation of the oxide particles in the outer layer of the oxide scale [17], which led to formation of an increased number of oxide particles. Further, the growth rate of the oxide particle is believed to be dependent on the diffusion rate of Fe ion in the inner oxide film according to the point defect model and dissolution-precipitation mechanism [30–32]. The diffusion of Fe ion in the inner oxide film, on the other hand, is dominated by grain boundary diffusion in accordance

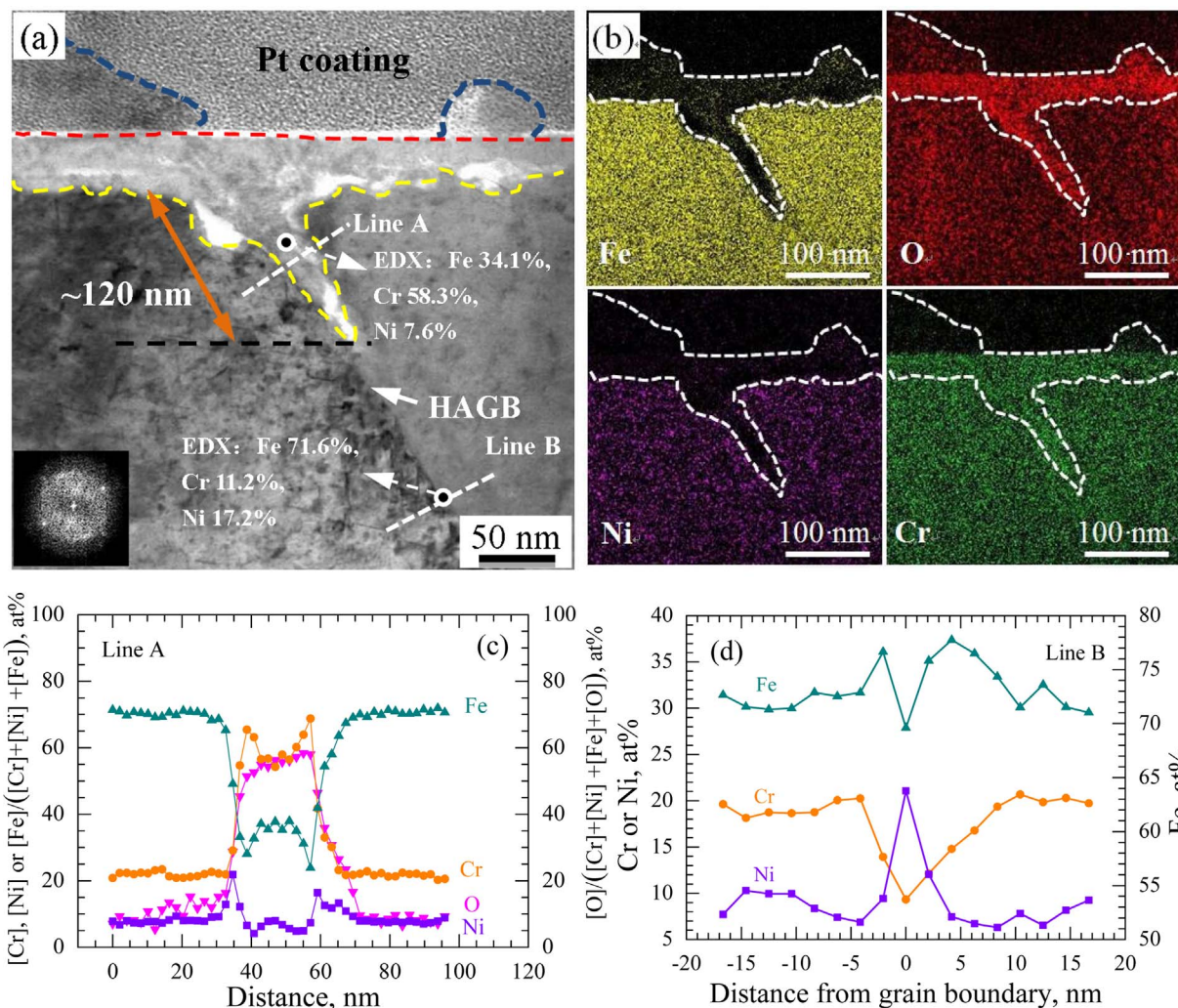


Fig. 11. (a), TEM observation of the localized corrosion at a grain boundary in 3-dpa irradiated 304NG SS. (b), Mappings for Cr, Fe, Ni, and O showing oxidation along the grain boundary. (c) and (d), EDX point-scans collected along lines A and B shown in (a), respectively.

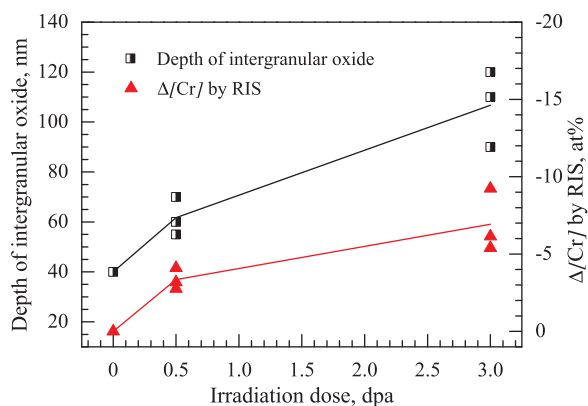


Fig. 12. Depth of intergranular oxide and the amount of Cr depletion ($\Delta[\text{Cr}]$) at the grain boundary by radiation-induced segregation as a function of irradiation dose.

with Robertson oxidation model [33–37]. The finer grains in the inner oxide film formed at a higher irradiation dose as confirmed by the high resolution TEM analysis (Fig. 8) could certainly promote the Fe ion diffusion through the grain boundary. As such, the growth of the oxide particle was promoted, resulting in a larger size of the oxide particle. In general, the increased size of the oxide particles on specimens with a higher irradiation dose resulted from a less protective nature of the

inner oxide film. On the other hand, the finer grains could also promote the inward diffusion rate of oxygen, leading to an increased thickness of the inner oxide.

The difference in grain size of the inner oxide formed on the SA and irradiated specimen was likely caused by the promoted diffusion of species by the high-density dislocation loops in the subsurface of the irradiated steel. Defects enhance the diffusion of the ions of oxygen and metal into the subsurface, accelerating the oxidation in these regions [19,38,39]. As such, a higher irradiation dose results in finer grain size of inner oxide by providing more sites of preferential oxidation.

4.2. Influence of irradiation on localized corrosion at grain boundary

The TEM observation and analysis as shown in Figs. 9–11 revealed difference in the intergranular corrosion behavior of the SA, 0.5-dpa and 3-dpa specimens. While the intergranular corrosion occurred at the grain boundary in both the SA and irradiated specimens, the depth of oxide penetration is larger for the irradiated specimens.

As has been acknowledged, a higher Cr content in an oxide film formed on austenitic alloys in high temperature water could improve the protectability of the oxide [26,27,40]. As such, the difference in the intergranular corrosion behavior of the SA and irradiated specimens is a combined effect of a change in the protectability of the inner oxide film and the Cr-depletion induced by RIS at the grain boundary. During exposure in the PWR primary water, a homogeneous and flat oxide

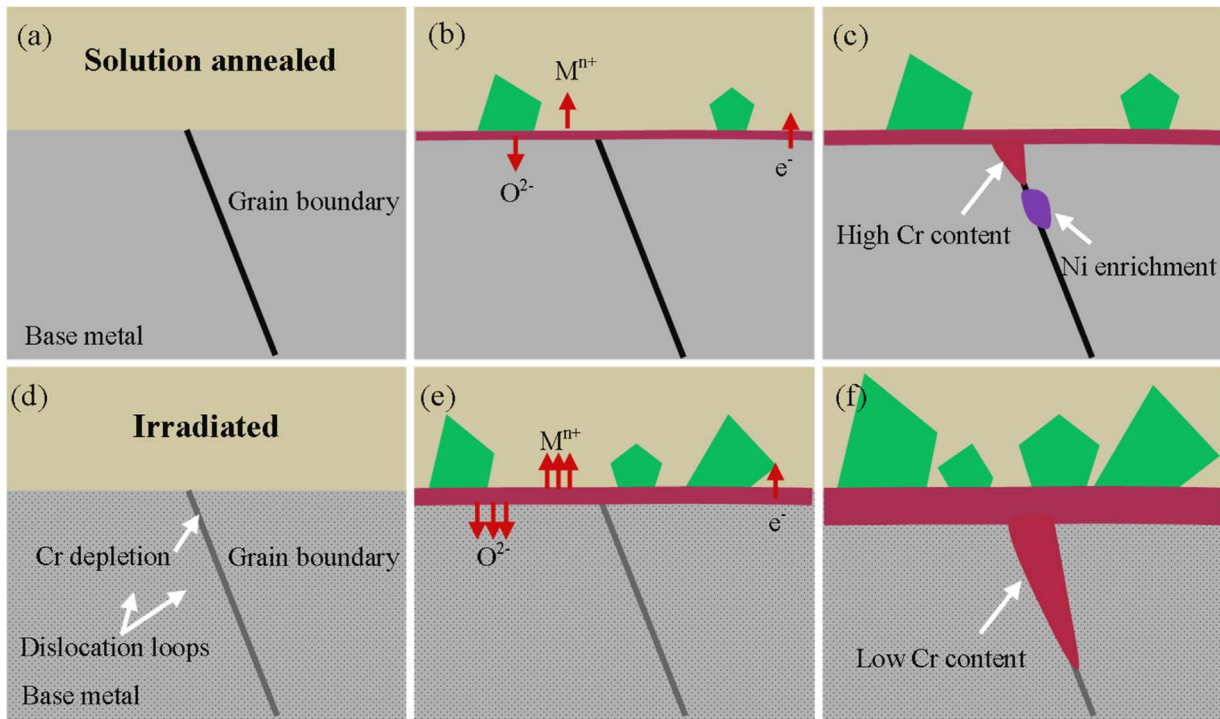
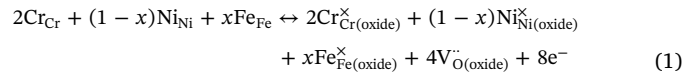


Fig. 13. Schematics showing the difference in the corrosion and intergranular corrosion behavior of (a–c) solution annealed and (d–f) irradiated specimens of 304NG SS in simulated PWR primary water.

scale was formed at first on the SA and irradiated specimens. As mentioned above, the inner oxide film of the oxide scale formed on the irradiated specimens was less protective, which allowed oxygen to penetrate it and oxidize the underneath base metal [41,42]. As such, intergranular corrosion on irradiated specimens beneath the surface oxide scale occurred more easily. In the following process of the intergranular corrosion, the Cr-depletion induced by RIS at the grain boundary could then promote further the intergranular corrosion by lowering the Cr content and the protectability of the oxide formed at the grain boundary. As shown in Fig. 12, the increase of intergranular oxidation depth with the irradiation dose is in correspondence with the increase of the magnitude of Cr depletion at the grain boundary. Since there is no Cr depletion at the grain boundary in the SA specimen, the oxide formed at the grain boundary had a higher Cr content and was more protective than that formed at the grain boundary in the irradiated specimens (shown in Figs. 9–11). As a result, a smaller depth of the intergranular oxidation was observed in the SA specimen. In addition, a localized enrichment of Ni was observed ahead the tip of the oxide at the grain boundary in the SA specimen, while no such Ni enrichment was formed in the irradiated specimens, shown in Figs. 9(b), 10(b) and 11(b). This also suggested a higher intergranular corrosion rate promoted by irradiation, since the Ni-enriched zone was attributed to a selective oxidation of Cr and Fe which removed Cr and Fe preferentially from the alloy [43–45]. The selective oxidation needs a long time, while a fast corrosion along the grain boundary could not provide sufficient time for the selective oxidation to occur [40,43].

The above discussion interprets the primary cause for the difference in the corrosion behavior of SA and irradiated specimens, in terms of the effect of the protectability of the oxide. The RIS, on the other hand, may also promote the intergranular corrosion by increasing the diffusion rate of metal cations at the oxide/metal interface. According to the point-defect mechanism for Fe-Cr-Ni alloys [27,30,46], the growth of oxide is determined by the oxygen vacancies formed at the oxide/metal interface by the following reaction:



where, the Cr_{Cr} , Ni_{Ni} , and Fe_{Fe} represent metal cation in cation site in the alloy, $\text{Cr}_{\text{Cr(oxide)}}^{\times}$, $\text{Ni}_{\text{Ni(oxide)}}^{\times}$, and $\text{Fe}_{\text{Fe(oxide)}}^{\times}$ refer to normal metal ion lattice sites in oxide, and $\text{V}_{\text{O(oxide)}}^{\bullet}$ represents oxygen ion vacancy in oxide. During the process of oxide growth, the metal cations were consumed at the oxide/metal interface and replenished by counter diffusion of metal vacancies [30]:



where, the m and M_{M} are metal atom and metal cation in cation site in the alloy ($m = \text{Fe}, \text{Cr}$ and Ni), respectively. $\text{V}_{\text{M}}^{\times'}$ represents metal vacancies surrounding the grain boundary.

Reactions (1) and (2) suggest that the formation of oxygen vacancy is limited by metal cations at the oxide/metal interface, and further determined by the amount of metal vacancy. As such, a change in the amount of metal vacancy near the grain boundary could affect the intergranular corrosion behavior. During irradiation, RIS occurred as a result of the unequal participation of solutes in the vacancy and interstitial fluxes to sinks [7,47]. Consequently, a higher magnitude of Cr-depletion at the grain boundary by RIS suggested an increased amount of metal vacancy, which could promote outward diffusion of metal cations to the oxide/metal interface. As a result, the enhanced intergranular corrosion on the irradiated specimen occurred by a combined effect of a faster diffusion rate of oxygen ions in oxide and metal cations at the oxide/metal interface.

The intergranular corrosion process for the SA and irradiated specimens as discussed above are schematically shown in Fig. 13.

The results of the current work provide insights into the mechanism of IASCC in austenitic SS in high temperature water. Since the corrosion/oxidation under stress favors the formation of cracking initiation, the promoted intergranular corrosion by irradiation suggests that a higher intergranular cracking initiation susceptibility in the highly irradiated steel can be expected [48].

5. Conclusions

Effect of irradiation on corrosion of 304NG SS in simulated PWR primary water was investigated. The following conclusions can be drawn from the investigation:

- 1 A higher irradiation dose generates a higher density and a larger size of dislocation loops, in addition to an increased degree of Cr depletion at the grain boundary.
- 2 The thickness of the oxide scale formed on 304NG SS is increased by a higher irradiation dose due to the promoted corrosion by irradiation induced defects.
- 3 Irradiation promotes intergranular corrosion of 304NG SS in simulated PWR primary water. This is attributed to the microstructural defects and depletion of Cr at the primary grain boundary induced by RIS.
- 4 Increasing irradiation dose favors formation of sites for IASCC initiation as a result of the enhanced localized corrosion at the grain boundary due to RIS.

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